

Login API Specification

Version: 1.0

Overview

This document specifies the Login API used by the front-end to authenticate users and obtain a JWT token.

Endpoint

- Path: `/api/v1/auth/login``
- Method: ``POST``
- Request Content-Type: ``application/json``

Request body:

```
{  
  "username": "string, required",  
  "password": "string, required"  
}
```

Responses:

- ``200`` — Successful authentication, returns ``{ "token": "..." }``
- ``400`` — Bad request (missing/invalid fields)
- ``401`` — Unauthorized (invalid credentials)

Acceptance Criteria

- Returns 200 and a valid JWT for correct credentials
- Returns 401 for incorrect credentials
- Returns 400 if required fields are missing

Mock Inputs

- `alice@example.com` / `CorrectHorseBatteryStaple`
- `bob@example.com` / `wrongpass`

Test Cases

1. TC-001 — Successful login
 - Preconditions: User `'alice@example.com'` exists with password `'CorrectHorseBatteryStaple'`
 - Steps: POST `/api/v1/auth/login` with JSON body
 - Expected: HTTP 200 with JSON `{ "token": "" }`
2. TC-002 — Invalid password
 - Preconditions: User `'bob@example.com'` exists
 - Steps: POST `/api/v1/auth/login` with JSON body
 - Expected: HTTP 401 Unauthorized
3. TC-003 — Missing username
 - Preconditions: none
 - Steps: POST `/api/v1/auth/login` with JSON body missing username
 - Expected: HTTP 400 Bad Request